Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	TeGrotenhuis, Ward Ph.D.		
Project Title:	Microchannel Phase Separations for Insitu Resource Utilization		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ward.tegrotenhuis@pnl.gov	**Fax:**	FY 509-372-1861
PI Organization Type:	INDUSTRY	**Phone:**	509-372-2268
Organization Name:	Battelle Northwest		
PI Address 1:	P.O. Box 999 MS K6-28		
PI Address 2:			
PI Web Page:			
City:	Richland	**State:**	WA
Zip Code:	99352	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	09/01/2000	**End Date:**	06/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stenkamp, Victoria (Battelle Northwest)		
Grant/Contract No.:	NAS3-00137		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Battelle is designing, building, and demonstrating compact, lightweight devices that can perform phase separations and gas separations in microgravity and reduced gravity environments. Initial work is being conducted at the Contractor's laboratory facilities to design, build, and test devices in order to establish the concepts, identify operating principles, and develop important scaling relationships. However, because of the limitations of normal gravity experiments, and the requirement for proof-of-concept in reduced gravity, ground-based reduced-gravity facilities, primarily the parabolic flight research aircraft, are being used. The objective of this effort is study both gas-liquid separation and partial condensation using microchannel devices in both normal and reduced gravity environments in order to develop this technology for In-Situ Resource Utilization (ISRU) for manned expeditions to Mars.		

	The purpose of this study is to achieve a better understanding and predictability of two-phase (gas-liquid) flow towards development of water recovery and purification systems that would be required as part of any in-situ resource utilization effort for the exploration of space. In addition, the ability to close the loop on consumables with such systems is necessary for long-term manned missions in space and would present significant cost savings in operating space-based facilities such as the International Space Station. The first phase of the effort involves the development of a two phase flow separator based on microchannel technology, followed by the development of a water condensor apparatus and scale-up of the technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)